BRAIN TUMOR CLASSIFICATION USING AI

Project Report

Project type	AI / Deep Learning Web Application
Interface	Streamlit interactive dashboard
Core model	TensorFlow / Keras trained model
Live demonstration	Open the deployed dashboard

This report summarizes the project's purpose, technical workflow, reported testing observations, deployment, and limitations.

Prepared for educational project review

1. Project Overview

Brain Tumor Classification Using AI is an educational deep-learning project demonstrating how a trained image-classification model can be connected to a web interface. A user uploads a brain MRI image, and the application displays the model's estimated probabilities for two configured classes: **Tumor** and **No Tumor**.

The project demonstrates an end-to-end workflow: loading a model, accepting an image, preprocessing it, running inference, and presenting a result through a browser-based dashboard.

2. Project Objectives

• Build a web interface for uploading MRI images.
• Load a trained TensorFlow/Keras model for image inference.
• Display a predicted class and model-estimated probabilities.
• Make the demonstration accessible through a deployed Streamlit URL.
• Practice model integration, testing, and responsible communication of AI limitations.

3. Technologies Used

Technology	Role
Python	Application logic and inference workflow
TensorFlow / Keras	Loading and running the trained neural-network model
Streamlit	Interactive dashboard and image-upload interface
NumPy	Numerical arrays and prediction-output handling
Pillow (PIL)	Opening and preparing uploaded images

4. Application Workflow

1. Upload: The user selects an MRI image in a supported format such as JPG, JPEG, or PNG.

2. Preprocess: The application opens the image and converts it into a numerical array. Image dimensions, color mode, and pixel scaling must match the model's training pipeline.

3. Predict: The prepared image is passed to the trained model, which returns an output interpreted as class probabilities.

4. Display: The dashboard shows the predicted class and estimated probabilities.

5. Review: Predictions should be compared against labelled test images and incorrect classifications investigated.

5. Model Testing and Current Observations

Earlier development notes reported these confusion-matrix counts: 400 correctly classified no-tumor images, 1,191 correctly classified tumor images, and 9 tumor images classified as no tumor. These counts imply 1,600 evaluated images, including 1,200 tumor cases. From these counts, overall accuracy is approximately **99.44%** (1,591/1,600), and tumor recall is approximately **99.25%** (1,191/1,200).

These figures were shared during development and have not been independently verified in this report. In later manual dashboard testing, an image selected from the Testing/notumor folder was displayed as “Tumor predicted” with a model-estimated tumor probability of 53.9%. This suggests the inference pipeline, class mapping, preprocessing, or model generalization requires further checking.

6. Known Issue and Recommended Debugging

Before describing the dashboard as reliable, verify the following:

• Confirm the exact class-to-index mapping used during training.
• Confirm whether the model output is one sigmoid value or a two-class softmax vector.
• Match image size, RGB/grayscale mode, and pixel normalization to training.
• Test several labelled images from both folders and record raw model outputs.
• Verify that the deployed app uses the latest app.py and intended model file.
• Recompute a confusion matrix on held-out data after correcting the inference pipeline.

7. Deployment and Access

The dashboard is deployed through Streamlit Community Cloud and is designed to open in a web browser. Live demonstration:

https://brain-tumor--classification-6dbh8vyhhmhcgudezxwbb7.streamlit.app/

For reproducibility, the GitHub repository should include app.py, requirements.txt, a README with setup instructions, and the trained model file if repository and hosting limits permit.

8. Limitations and Responsible Use

This is an educational demonstration only. It is not a medical device and has not been clinically validated. A model output must not be used to diagnose or rule out a brain tumor or to guide treatment. Medical interpretation requires qualified healthcare professionals and appropriate clinical evaluation.

Further work could include verifying data splits and class labels, evaluating sensitivity and specificity on independent data, testing robustness to image-acquisition differences, and documenting the training pipeline.

9. Conclusion

The project demonstrates how a trained image-classification model can be integrated into an interactive Streamlit dashboard. It provides a foundation for learning about preprocessing, inference, deployment, and evaluation. The priority for further development is to resolve the mismatch observed during manual testing and validate the full inference pipeline.

Note: Performance figures are reported from development notes and should be independently re-evaluated. This report does not claim clinical accuracy.

Live demo: Open dashboard